AfriGen-D Beacon — Honours Project Catalogue

Sixteen projects spanning privacy research, federation, clinical extension, and new Beacon variants

AfriGen-D · UCT CBIO

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Contents

Overview

This document catalogues sixteen honours-level research projects built on the AfriGen-D GA4GH Beacon v2 deployment. Each project is scoped for a 4–6 month honours dissertation, ties to the existing codebase or live service, and produces both an engineering deliverable and a measurable research result.

How the catalogue is structured

Projects are grouped into seven themes:

- A. Privacy & security — attacks and defenses on Beacon responses
- B. Federation & networking — the African Beacon Network and its operations
- C. Standards & conformance — Beacon v2 specification testing
- D. Clinical & phenotype queries — Phenopackets, PharmGKB integration
- E. UX & visualisation — privacy-aware result presentation
- F. Infrastructure & data engineering — ingest, indexing, scaling
- G. Other Beacon types — pathogen, methylation, cancer, plant, microbiome, imaging

Each project entry contains six fields:

Field	What it captures
Research question	The thesis-level question the work answers
User-facing question	What end users can now ask of the system
Utility	Why it matters (to AfriGen-D, to the field, to the student)
Skills learned	Technical and research skills the student gains
Knowledge needed	Prerequisite background
Difficulty / months / risk	Calibration for a typical honours student

Why these projects, on this Beacon

The AfriGen-D Beacon is the only currently deployed, verifier-conformant GA4GH Beacon v2 over an African reference cohort (V6HC-S_AFR; 1,895 individuals on GRCh38). Its query log is being captured. Its codebase is documented (`CLAUDE.md`, `docs/SPEC_CONFORMANCE.md`). Its sister network aggregator (African Beacon Network) is in active development.

Most of the published Beacon literature evaluates on European 1000 Genomes data. Many open questions — privacy attacks across populations, federated latency, non-variant Beacons for African endemic disease — are open precisely *because* nobody has had a real African Beacon to evaluate against. That is the gap this catalogue exploits.

Theme A — Privacy & security

A1. Replicating beacon re-identification attacks against an African reference beacon

Research question. Does the membership-inference attack power on a Beacon vary with the ancestry of the underlying cohort? Are African-ancestry Beacons more, less, or equally vulnerable than European-ancestry Beacons given matched sample sizes? Which feature of the allele-frequency spectrum (rare-variant load, LD block size, F_{ST}) drives the difference, if any?

User-facing question. "If I run a Beacon over my 500 South African individuals, how many queries does an attacker need to recover one of them? Should I require authentication for queries below MAF X?"

Utility. AfriGen-D operates a public Beacon over African genomes and needs to know empirically whether the published European attack-power numbers transfer. If they do, the privacy posture is weaker than assumed. If they don't, that is a defensible publication arguing for population-aware risk modelling. Every privacy paper since 2015 evaluates on European 1000G data; whether attacks scale across populations is genuinely unknown.

Skills learned.

- *Technical:* Python statistical computing (scipy, numpy, pandas), VCF parsing (pysam/cyvcf2), API client design with rate-limit handling, bootstrapping/Monte Carlo simulation, plotting (matplotlib/seaborn).
- *Research:* threat modelling, re-implementing methods from a paper, designing power experiments, writing about negative/null results honestly.

Knowledge needed. Solid statistics (likelihood-ratio tests, hypothesis testing) — required. Basic population genetics (allele frequency, MAF, LD) — self-teachable in 3 weeks. Python proficiency — required. Familiarity with HTTP APIs.

Difficulty. 7/10 (medium-high). The math is well-explained in Shringarpure & Bustamante (2015) but the student must implement it. The novelty (African comparison) is what makes it honours-publishable.

Timeline. 4–5 months. Month 1: literature + setup + 1000G European replication. Month 2: African-data attack. Month 3: comparative power analysis. Month 4: writeup + paper draft.

Risk. Low. Worst case: attack works the same on Africans –> still a publishable null result.

A2. Differential-privacy noise layer for Beacon responses

Research question. Can we provide ε-DP guarantees on Beacon boolean responses while keeping researcher utility above a useful threshold? What is the Pareto frontier of (ε, utility, attack power) on real African data?

User-facing question. “If I set $\varepsilon = 1.0$, what fraction of legitimate variant lookups will I now answer wrongly? What’s the minimum cohort size I need before DP becomes ‘free’ (utility loss $< 5\%$)?”

Utility. This is a deployable security feature — the prototype hardens the public Beacon directly. DP for genomic Beacons is well-studied theoretically but few production-deployed implementations exist; a working open-source middleware would be cited. Differential privacy is one of the most sought-after specialisations in industry (Apple, Google, US Census).

Skills learned.

- *Technical:* Django middleware development, Python decorators, MongoDB query interception, statistical mechanism design, A/B testing, utility benchmarking.
- *Research:* reading formal cryptography/privacy papers, ε-DP composition reasoning, designing utility metrics, defending design choices in writing.

Knowledge needed. Strong stats and probability — required. Basic understanding of DP (ε, δ, sensitivity, Laplace mechanism) — acquirable in 4–6 weeks from Dwork & Roth’s *Algorithmic Foundations of DP*. Django + OOP required.

Difficulty. 8.5/10 (hard). The most theoretically demanding project on the list. Best assigned to a strong honours student aiming at MSc, or an MSc-bound student. Do not give to a weak student — they will produce something that *looks* DP but is mathematically broken.

Timeline. 5–6 months. Realistically tight; many honours students under-budget the theory phase.

Risk. Medium. Failure mode: implements something they call DP but is broken. Mitigation: require a formal proof sketch reviewed by a co-supervisor with cryptography background.

A3. Defending against the 2025 Beacon Reconstruction Attack

Research question. Can server-side query-budgeting plus allele-frequency suppression reduce reconstruction-attack F1 below 0.5 without breaking benign-user workloads? Which countermeasure subset gives the best security/utility ratio?

User-facing question. “If I’m a benign researcher running 1,000 queries/day, will this defense affect me? How many SNPs of a member’s genome could a determined attacker still reconstruct under our hardened policy?”

Utility. The 2025 reconstruction attack is brand new and the AfriGen-D Beacon is genuinely vulnerable. There are no published deployable defenses yet. First-mover advantage; a working defense paper would be timely and citation-attractive.

Skills learned.

- *Technical:* same as A2 (middleware + statistics) plus high-dimensional data analysis (correlation matrices, low-rank approximation), Redis/cache design for query budgets.
- *Research:* constructive defense design (harder than offense), trade-off curves, evaluating against an adversary you simulate yourself.

Knowledge needed. Linear algebra (correlation matrices, low-rank approximation) required. Comfortable reading recent ML/security papers. Understanding of LD and haplotype structure helps; can be acquired during the project.

Difficulty. 8/10 (hard). Slightly less mathematically demanding than A2 (fewer formal proofs) but more engineering surface area. Better suited than A2 for an engineering-strong, theory-medium student.

Timeline. 5–6 months. Phase 1: replicate attack (1.5 months). Phase 2: design defenses (2 months). Phase 3: evaluation (1.5 months). Phase 4: writeup.

Risk. Medium-high. Failure mode: defenses break utility too much to be usable. Mitigation: define "usable" up front via a baseline researcher use case.

Theme B — Federation & networking

B1. Beacon network anomaly detection / abuse classifier

Research question. Can supervised ML on query logs distinguish privacy-attack traffic from researcher traffic at AUC > 0.95 with < 1% false-positive rate? Which features (query inter-arrival, MAF distribution, position diversity) carry the most signal?

User-facing question (operator-facing). "Is this IP currently running a Shringarpure–Bustamante attack? How much abuse traffic did we receive this month?" (Grafana panel.)

Utility. The `query_logs` collection is already populated but not analysed. This converts a passive log into an active defense layer. End-to-end ML system (feature engineering –> model –> deployment –> monitoring) — a highly employable pattern.

Skills learned.

- *Technical:* MongoDB aggregation pipelines, feature engineering on time-series logs, scikit-learn (logistic regression, gradient boosting, isolation forest), model evaluation (precision/recall/AUC), Grafana panel design.
- *Research:* building synthetic adversarial datasets, dealing with extreme class imbalance, threshold selection, false-positive cost reasoning.

Knowledge needed. Intro ML course (decision trees, regression, basic evaluation metrics) — required. Python data science (pandas, sklearn) — required. Basic SQL/NoSQL — pickup in week 1.

Difficulty. 5.5/10 (medium). Most students can complete this. The hard part is *evaluation* — getting realistic positive examples — not the modelling itself. Pairs naturally with A1.

Timeline. 4 months. Month 1: log analysis + feature design. Month 2: synthetic attack generation. Month 3: training + tuning. Month 4: deployment + dashboard + writeup.

Risk. Low.

B2. Latency benchmarking & SLO design for the African Beacon Network

Research question. What is the latency cost of federation as a function of node count N? Which architectural change (parallel fan-out, partial- result streaming, response-cache) yields the largest p95 reduction per engineering hour?

User-facing question. "If I query 12 African Beacons, how long should I wait before retrying? What SLO can the network credibly promise to clinical users?"

Utility. The aggregator (`afrigend-beacon-network`) is in active development; its real performance characteristics are unknown. This produces actionable tuning advice. Benchmarking studies of federated genomic systems are surprisingly rare; most papers handwave performance.

Skills learned.

- *Technical:* OpenTelemetry / distributed tracing, Locust load-testing, async Python (asyncio, aiohttp), Prometheus / Grafana, network profiling (tcpdump, mtr), connection pooling.
- *Research:* designing controlled experiments on networked systems, latency-distribution analysis (p50/p95/p99), drawing meaningful conclusions from noisy measurements.

Knowledge needed. Comfortable with Linux/Docker — required. Networking fundamentals (HTTP, DNS, TLS basics) — strongly recommended. Async Python — self-teach in 2 weeks.

Difficulty. 5/10 (medium). Lots of bounded sub-tasks; a methodical student will do well.

Timeline. 4 months. Month 1: tracing setup + initial measurements. Month 2: load testing. Month 3: optimisation experiments. Month 4: SLO recommendation + writeup.

Risk. Low.

Theme C — Standards & conformance

C1. Property-based fuzzer for Beacon v2 implementations

Research question. How many GA4GH Beacon v2 spec violations exist in publicly deployed Beacons? Which spec sections produce the highest violation density (i.e., which parts of the spec are ambiguous)?

User-facing question. "Is my Beacon implementation actually compliant?" (developer tool output.) "Which spec sections need clarifying language?" (input to the GA4GH working group.)

Utility. AfriGen-D would own the only open-source Beacon-v2-specific fuzzer. Discovers real bugs in real production beacons –> publishable as a tool paper plus a "we found *N* bugs in *N* implementations" empirical study. Property-based testing is high-signal on a CV.

Skills learned.

- *Technical:* Hypothesis library (Python), JSON schema, OpenAPI tooling, test-oracle design, structured bug reporting.
- *Research:* spec-driven thinking, systematic exploration vs example-driven testing, ethics of bug disclosure.

Knowledge needed. Python + good unit-testing intuition — required. Familiarity with REST APIs and JSON — required. Reading formal specs without giving up — partly a personality fit.

Difficulty. 5/10 (medium). Mechanically not hard; the difficulty is patience and attention to detail. Best fit for a meticulous student.

Timeline. 4 months.

Risk. Low. Worst case: tool produces clean Beacons with no bugs –> student writes “limitations and false-negative analysis” and still has a thesis.

Theme D — Clinical & phenotype queries

D1. Phenopackets-driven cohort discovery on the AfriGen Beacon

Research question. What is the smallest schema subset needed to answer 80% of clinically meaningful queries on African cohort data? Where do phenotype representations diverge across H3Africa cohorts, and what mapping rules harmonise them?

User-facing question (clinician/researcher). “How many sickle-cell carriers in our cohorts also have variant X in haemoglobin gene Y? Are there >= 5 individuals with HbF persistence and a *BCL11A* enhancer variant? Which African cohorts contain pediatric individuals with documented G6PD deficiency?”

Utility. Highest user-facing impact in the catalogue. The current Beacon answers “is variant X present?” but cannot answer “how many sickle-cell carriers in cohort Y also have variant X?” — the question clinicians actually ask. Closing this gap turns the Beacon from a demo into a clinical tool. African pharmacogenomics and rare-disease cohorts are critically under-discoverable.

Skills learned.

- *Technical:* MongoEngine ODM design, JSON schema validation, ontology lookup (HPO, OMIM, Mondo), DRF view design, end-to-end testing.
- *Research:* schema design trade-offs, ontology mapping, evaluation against clinical query exemplars.

Knowledge needed. Python + Django basics (or willingness to learn — the existing codebase is a strong example). Some genetics/genomics background. JSON / REST literacy — required.

Difficulty. 6/10 (medium). Well-bounded and high-impact. Lots of sub-tasks of varying difficulty, so good for honours: student can always retreat to a smaller scope if blocked.

Timeline. 5 months.

Risk. Low. Even partial implementation (individuals only, no biosamples) is useful and thesis-able.

D2. Pharmacogenomics-aware Beacon extension (AGMP integration)

Research question. Can a star-allele-aware Beacon respond to clinically meaningful PGx questions faster than ad-hoc SQL on PharmCAT outputs? Do African star-allele frequencies inferred from the Beacon agree with published PharmGKB Africa-specific data?

User-facing question (clinician). “How prevalent is *CYP2D6*17* in our Southern African cohort? Is there at least one individual with the *CYP2B6*6/*6* genotype (relevant to efavirenz dosing)? Which PGx-relevant variants are observed in our HIV cohort?”

Utility. AGMP is already a sister site of AfriGen-D; tighter integration into the Beacon is a stated AfriGen-D goal. Pharmacogenomic Beacons are not a standard yet — the project would help define the pattern.

Skills learned.

- *Technical:* CYP / star-allele nomenclature, PharmGKB and CPIC data, REST API extension, frontend integration (Next.js / TypeScript if doing UI).
- *Research:* pharmacogenomic biology, clinical relevance reasoning.

Knowledge needed. Python + Django — required. Some pharmacology / genetics interest — required for motivation.

Difficulty. 4.5/10 (medium-easy). Smaller research question; mostly engineering. Good fit for a strong-engineer / weak-statistician student.

Timeline. 4 months.

Risk. Low.

Theme E — UX & visualisation

E1. Privacy-aware visualisation of Beacon results

Research question. Do researchers' decisions change when shown banded counts vs raw counts vs DP-noised counts? Which visualisation form yields the best researcher-trust score given equivalent privacy guarantees?

User-facing question. "Should I bother running a follow-up study on this variant — how rare is rare? Is the data here statistically powered for my downstream analysis?"

Utility. The current frontend (`DatasetResults.tsx`) shows raw counts — which the privacy literature shows is risky. This project produces evidence-backed redesigns. HCI-meets-privacy in genomics is genuinely understudied.

Skills learned.

- *Technical:* React + TypeScript, Next.js App Router, Tailwind CSS, accessibility basics, user-study tooling.
- *Research:* designing user studies, ethics-board applications, interview coding, qualitative + quantitative analysis.

Knowledge needed. TypeScript or willingness to learn — required for implementation. An HCI or research-methods course is a strong plus.

Difficulty. 5.5/10 (medium). Difficulty balanced toward research design rather than coding. Best for a student with HCI inclination, possibly from a non-CS background.

Timeline. 4 months. Note ethics-board approval can eat 2–6 weeks — start early.

Risk. Medium. The ethics timeline is the biggest single risk. Mitigation: backup task (technical-only redesign without user study).

Theme F — Infrastructure & data engineering

F1. Bulk-ingest pipeline benchmarking on V7HC-S panel

Research question. What is the throughput ceiling of the current Mongo-based Beacon backend at v7 reference-panel scale? Does an analytical store (ClickHouse / DuckDB+Parquet) outperform the document store on Beacon-typical workloads?

User-facing question. “Can we host V7HC-S in production without changing the storage layer? What’s the cost-per-query at v7 scale?”

Utility. V7HC-S exists on ILIFU but isn’t loaded into the Beacon. Whether the current pipeline can ingest it at scale is unknown. A benchmark answers the question and produces tuning recommendations. Data engineering skills are universally hireable.

Skills learned.

- *Technical:* Nextflow pipeline development, MongoDB indexing and sharding, comparative database evaluation (Mongo vs ClickHouse vs DuckDB+Parquet), profiling tools (cProfile, py-spy), benchmarking methodology.
- *Research:* designing fair comparisons across heterogeneous systems, choosing evaluation metrics that survive scrutiny.

Knowledge needed. Linux + Docker — required. SQL or basic database concepts — required. Nextflow can be self-taught.

Difficulty. 5.5/10 (medium). Lots of moving parts; a methodical student will produce a strong report.

Timeline. 4 months.

Risk. Low-medium. ILIFU access scheduling can be a bottleneck; ensure the student has compute-node SSH access from week 1.

Theme G — Other Beacon types

G1. Pathogen Beacon for African endemic diseases (TB / HIV / malaria / mpox)

Research question. Can a privacy-preserving Beacon over pathogen sequence data support outbreak-response queries with sub-second latency? What is the minimum spatio-temporal coarsening that prevents source-case re-identification while retaining epidemiological utility?

User-facing question (epidemiologist / public-health official). “Has the *katG S315T* isoniazid-resistance mutation been observed in TB isolates from Western Cape in the last 90 days? Which African countries have reported the *Plasmodium falciparum* K13 C580Y artemisinin-resistance mutation since 2025? Is this newly-sequenced HIV pol fragment consistent with strains already in our network?”

Utility. Africa generates pathogen sequence data continuously (NICD, KEMRI, ACEGID, CERI), but discovery infrastructure is fragmented. Viral Beacon (CRG) only covers SARS-CoV-2 with European compute. There is *no* deployed Beacon for HIV, TB, malaria, or mpox — first-mover territory.

Skills learned.

- *Technical:* phylogenetics tooling (Nextstrain auspice JSON, pangolin lineage, USHER), pathogen-specific bioinformatics (HIV drug-resistance interpretation, MTB lineage typing, *Plasmodium* drug-resistance markers), GISAID/NCBI data ingestion, Django modelling.
- *Research:* outbreak data ethics, surveillance vs research data distinction, geospatial privacy, time-binning trade-offs.

Knowledge needed. Bioinformatics fundamentals (alignments, FASTA, mutation calling) — required. Some interest in microbiology / epidemiology. Python + Django.

Difficulty. 7/10 (medium-high). The pathogen-specific bioinformatics is the real challenge; the Beacon shape is the easy part.

Timeline. 5 months. Pick *one* pathogen — TB is concrete and politically valuable; HIV is data-rich; malaria is clinically urgent.

Risk. Medium. Data access can be slow (DUAs from data providers). Mitigation: start with public NCBI/GISAID data only.

G2. Methylation Beacon (MBeacon) for African epigenomes

Research question. Does the Hagestedt et al. (2019) SVT2 mechanism preserve utility on African methylation cohorts of realistic size? Are African EPIC array cohorts more or less vulnerable to the methylation membership-inference attack than European ones?

User-facing question (epigenomics researcher). "Is hypermethylation at *MGMT* observed in our pediatric ALL cohort? Which African studies report differential methylation at *AHRR* in tobacco-exposed individuals?"

Utility. Methylation is increasingly recognised as a key intermediate phenotype linking environment and disease in African populations. There is no African epigenetics discovery infrastructure. MBeacon was published in 2019 but never productionised; rebuilding it on Beacon v2 + deploying on real data closes a 7-year gap in the literature.

Skills learned.

- *Technical:* Illumina 450K/EPIC methylation data, β-value normalisation, R/Bioconductor (minfi), DP mechanisms (sparse vector technique), Django modelling.
- *Research:* re-implementing a privacy mechanism from a security venue paper, comparing utility curves to original work.

Knowledge needed. Statistics + comfort with continuous data (β-values are 0–1 floats, not discrete). Strong ML/probability fundamentals.

Difficulty. 8/10 (hard). Combines the rigour of A2 with new biology.

Timeline. 5–6 months.

Risk. Medium. Sourcing African methylation data is non-trivial. Mitigation: start with TCGA / GEO public data, validate, then approach H3Africa cohorts.

G3. African cancer / CNV Beacon (Progenetix-style on local data)

Research question. Are CNV profiles of common cancers (breast, cervical, oesophageal) in African cohorts distinguishable from published TCGA / Progenetix profiles? Does Beacon-style discovery accelerate cohort assembly for African cancer studies?

User-facing question (cancer researcher). "How many oesophageal squamous-cell carcinomas with 11q13 amplification are available in African biobanks? Which African cancer studies report 8q24 gain in triple-negative breast cancer? Are there >= 30 African cervical cancer samples with HPV16 integration plus 3q26 amplification I could request?"

Utility. African cancer genomics is severely under-represented in databases like Progenetix (mostly European/Asian samples). A locally hosted, Beacon-discoverable repository is a real public good. Hangjia Zhao & Michael Baudis (UZH) maintain Progenetix and would likely collaborate.

Skills learned.

- *Technical:* CNV calling and representation, ICD-O ontology, Plotly/D3 chromosome plots, MongoDB range queries (CNVs are intervals).
- *Research:* cancer ontology mapping, somatic-vs-germline distinction (and why it matters for privacy), evaluating against Progenetix as gold standard.

Knowledge needed. Some cancer genomics interest — required for motivation. Python + Django + interval/range query thinking.

Difficulty. 6/10 (medium). Well-bounded; lots of guidance from Progenetix docs.

Timeline. 5 months.

Risk. Low-medium. Data access (need African cancer CNV calls) is the main uncertainty.

G4. African crop / agricultural Beacon (cassava / sorghum / teff)

Research question. What schema extensions does Beacon v2 require to support germplasm metadata (MCPD passport descriptors) cleanly? Does Beacon-style discovery improve the findability of African crop pangenome variants relative to BLAST-and-FTP workflows?

User-facing question (plant breeder / crop geneticist). "Which cassava accessions in the African pangenome carry the *MePSY1* allele linked to provitamin-A? Are there sorghum lines in African collections with the *Sb04g005530* drought-tolerance haplotype? Which fonio accessions show structural variation in agronomically-relevant genes?"

Utility. Although AfriGen-D is human-genomics-focused, the African agricultural genomics community (IITA, Africa Rice, ICRISAT, AOCC) has substantial sequencing data and zero discovery infrastructure. Bridging human and crop genomics is strategically valuable for funding. GA4GH Beacon documentation explicitly mentions plants but no public plant Beacon exists.

Skills learned.

- *Technical:* plant genomics conventions, pangenome data formats (VG, GraphAligner outputs), MCPD passport-data standard, Django data modelling.
- *Research:* designing standard extensions, engaging with the GA4GH Plant data working group.

Knowledge needed. Plant biology / agronomy interest — strongly preferred. Python + Django.

Difficulty. 6/10 (medium). Difficulty mostly in the data (plant genomes are big and weird) not the Beacon part.

Timeline. 5 months.

Risk. Medium. ILIFU has compute capacity for a small pangenome. Risk of scope creep into pangenome graph queries (defer to MSc).

G5. Microbiome Beacon for African gut/oral microbiome cohorts

Research question. Can a Beacon abstraction support cross-cohort microbiome discovery despite OTU/ASV/MAG heterogeneity? What metadata schema enables meaningful federated microbiome queries across African gut-microbiome cohorts?

User-facing question (microbiome researcher). "Which African cohorts contain *Prevotella copri* at relative abundance >= 5%? Are there cohorts with both stunted children and reduced *Bifidobacterium* abundance? Which functional pathways (e.g., bile-acid metabolism) are observed in cohorts from rural Kenya vs urban Cape Town?"

Utility. H3Africa includes microbiome studies — a Beacon for "is taxon X / function Y observed in cohort Z" would be unique globally. Microbiome data sharing is famously chaotic; a discovery layer is sorely needed.

Skills learned.

- *Technical:* QIIME2 / DADA2 pipelines, taxonomy databases (SILVA, GTDB), functional profiling (HUMAnN, PICRUSt2), Django data modelling.
- *Research:* schema design in an unstandardised area, microbial community ecology, data-sharing ethics for stool samples.

Knowledge needed. Microbiology / ecology background — required. Python + R bioinformatics.

Difficulty. 7/10 (medium-high). Schema design in an unstandardised domain is the hard part — needs taste, not just code.

Timeline. 5 months.

Risk. Medium-high. Schema-design projects can drift; requires a hands-on co-supervisor with microbiome experience.

G6. Imaging Beacon for African medical imaging cohorts

Research question. Can DICOM metadata-only queries support cohort assembly for medical-AI training without exposing pixel data? Which subset of RadLex/SNOMED-CT findings produces the most useful Beacon discovery vocabulary for African TB and diabetic-retinopathy cohorts?

User-facing question (medical-AI researcher). "How many adult chest X-rays with confirmed pulmonary TB and HIV co-infection are discoverable across African hospitals? Are there >= 1,000 retinal images with proliferative diabetic retinopathy from African populations? Which sites have paired CT/MRI imaging for stroke patients with neurogenetic-testing data?"

Utility. H3Africa and CIDRI-Africa have large TB X-ray and diabetic-retinopathy image datasets that are not discoverable. An Imaging Beacon would be high-impact for AI-for-Africa research. No GA4GH-compliant Imaging Beacon exists; bridging GA4GH with federated-imaging work (Kaapana, MONAI Federated) is novel.

Skills learned.

- *Technical:* DICOM / DICOMweb, medical-imaging metadata, RadLex / SNOMED-CT, Django.
- *Research:* the privacy boundary between metadata (safe) and pixels (high-risk); designing metadata-only protocols.

Knowledge needed. Some medical imaging exposure — strongly recommended. Python + Django.

Difficulty. 7/10 (medium-high). Cross-domain — student must learn DICOM as a sub-project.

Timeline. 5 months.

Risk. Medium. Hospital / IRB politics around imaging data. Mitigation: start with public TB CXR datasets (Shenzhen, Montgomery) before approaching local cohorts.

Difficulty / fit summary table

#	Project	Difficulty	Months	African novelty	Publishable?
A1	Re-id attacks on African Beacon	7/10	4–5	***	Likely
A2	DP noise layer	8.5/10	5–6	**	Yes
A3	Reconstruction-attack defense	8/10	5–6	**	Yes (high impact)
B1	Anomaly detection	5.5/10	4	*	Workshop
B2	Latency benchmarking	5/10	4	*	Tool paper
C1	Property-based fuzzer	5/10	4	*	Tool paper
D1	Phenopackets cohort discovery	6/10	5	**	Cite-able
D2	PharmGKB extension	4.5/10	4	**	Domain venue
E1	Privacy-aware visualisation	5.5/10	4	*	HCI venue
F1	V7 ingest benchmarking	5.5/10	4	*	Workshop

#	Project	Difficulty	Months	African novelty	Publishable?
G1	Pathogen Beacon (TB / HIV / malaria)	7/10	5	****	Yes (high impact)
G2	MBeacon for African epigenomes	8/10	5–6	***	Yes
G3	African cancer / CNV Beacon	6/10	5	***	Cite-able
G4	Crop / agricultural Beacon	6/10	5	****	First-mover
G5	Microbiome Beacon	7/10	5	****	First-mover
G6	Imaging Beacon (TB CXR / DR)	7/10	5	****	First-mover

Cohort designs

If you can run only three students simultaneously, four sensible combinations:

Africa-impact cohort. G1 (Pathogen) + G4 (Crop) + D1 (Phenopackets). Maximum public-good impact, three under-served domains, complementary skills.

Privacy-research cohort. A1 (Re-id attacks) + A3 (Reconstruction defense) + G2 (MBeacon). A coherent privacy pipeline; A1's attack output feeds A3.

Infrastructure / standards cohort. B2 (Latency) + C1 (Fuzzer) + F1 (V7 ingest). All three improve operational capacity; cleanest non-overlapping engineering.

Mixed showcase cohort. G1 (Pathogen) + A1 (Re-id) + D1 (Phenopackets). One of each kind — domain extension, privacy research, clinical utility. Best for diverse student strengths.

References

Live deployments (AfriGen-D)

The Beacons students will actually query as part of their projects:

- ARDI Beacon — https://beacon.ardi.africa/
- AfriGen-D Beacon — https://beacon.afrigen-d.org/
- African Beacon Network (federation aggregator) — https://beacon-network-dev.afrigen-d.dev/

Foundational papers

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- Rueda et al. (2022), *Beacon v2 Reference Implementation: a toolkit to enable federated sharing of genomic and phenotypic data*. Bioinformatics 38(19): 4656–4657. DOI: 10.1093/bioinformatics/btac568.
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Privacy attacks

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- Yilmaz et al. (2025), *Beacon Reconstruction Attack: reconstruction of genomes in genomic data-sharing beacons using summary statistics*. Bioinformatics 41(6): btaf273.

Beacon variants

- Hagestedt et al. (2019), *MBeacon: privacy-preserving beacons for DNA methylation data*. NDSS Symposium 2019.
- Huang et al. (2025), *Privacy-preserving framework for genomic computations via multi-key homomorphic encryption*. Bioinformatics 41(3): btae754.
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Web resources

- GA4GH Beacon: https://www.ga4gh.org/product/beacon-api/
- Beacon Project: https://genomebeacons.org/
- Beacon docs: https://docs.genomebeacons.org/
- Beacon spec: https://github.com/ga4gh-beacon/specification
- Twelve quick tips for deploying a Beacon (2024): https://genomebeacons.org/publications/2024-03-01-Beacon-Tips/
- Progenetix (cancer/CNV Beacon): https://docs.progenetix.org/
- Viral Beacon: https://inb-elixir.es/news/viral-beacon-beacon-ocean-sars-cov-2-data
- AfriGen-D: https://afrigen-d.org/
- AGMP: https://agmp.afrigen-d.org/